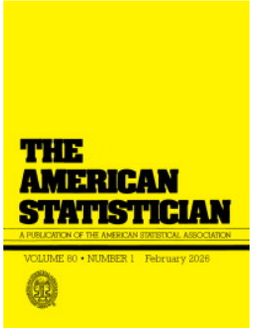

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Explainable Rank-Based Anomaly Detection

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Abstract

This paper introduces a practical and flexible framework for anomaly detection in settings characterized by multiple features. It presents a novel application of the sum of absolute rank differences, which considers different characteristics across multiple groups. The approach combines anomaly identification with explainable diagnostics through the development of comparative curves and a feature importance metric, providing explainability for unsupervised anomaly detection. This method provides graphical visualizations to highlight anomalous behaviors and identify the influential features contributing to these anomalies. The utility and flexibility of the proposed method are demonstrated through real-world applications in domains such as healthcare billing and review ranking, showcasing its effectiveness in identifying and explaining anomalies across different settings.

Keywords: Anomaly Detection; Rank-Based Methods; Feature Importance; Explainable Unsupervised Model

1 Introduction

Anomaly and outlier detection methods play a crucial role across various application domains, serving to identify abnormal data behavior and uncover patterns and deviations (Hodge and Austin, 2004; Ben-Gal, 2005; Wang et al., 2019). Multi-attribute outlier and anomaly detection have been explored from different angles (Rousseeuw and Van Zomeren, 1990; Filzmoser et al., 2004; Hubert et al., 2005; Kannan and Manoj, 2015), including classical methods such as Mahalanobis distance and Cook's distance (Cook, 1977), among others. While statistical outlier detection methods mostly rely on sampling models to be tested against the presence of outliers, anomalies are broadly defined as patterns that do not conform to expected normal behavior (Chandola et al., 2009). Many practical applications concerning anomaly detection occur in different domains such as fraud detection, cyber-intrusion detection, healthcare (Huang et al., 2013). Most of these methods rely on numerical differences to discern anomalies. However, there are scenarios where utilizing ranks can offer advantages. For instance, consider a medical provider with prescription percentages that are reversed compared to a benchmark, albeit with minor numerical differences. While this could be of interest to auditors due to reverse ranks,

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existing methods may not detect this as an anomaly. This manuscript introduces a straightforward yet innovative and explainable rank-based anomaly detection approach. Our method involves computing the sum of absolute rank differences, comparing them with a benchmark, and generating curves to highlight anomalous behavior. Additionally, we introduce a feature importance metric to quantify the contribution of each feature to the observed anomalies. The use of these metrics, along with graphical visualizations of anomalous data, are illustrated through real-world applications in hotel reviews and health care billing.

Ranking holds significance across a myriad of domains and real-world scenarios encompassing health economics, competition dynamics, and social preferences (Taplin, 1997). Detecting observations with specific ranks amidst multiple attributes poses challenges, evident in contexts such as healthcare economic resource allocation (Hernández Alava et al., 2013) and political decision-making (Arrow, 2012). Whereas, rank-based methodologies find application in various statistical contexts, including rank-based estimators (Kloke and McKean, 2012), Wilcoxon rank-sum test, Spearman's rank-order correlation, rank-based regression (Jureckova, 1971), information retrieval (Sejr and Schneider-Kamp, 2021), and recommendation models (Schedl and Bauer, 2017). Several data mining algorithms include rank-based evaluation of multiple characteristics (Pihur et al., 2007; Santus et al., 2018; Calders et al., 2006). For instance, Hawkins et al. (2002) assess the performance of the replicator neural networks using a ranked score measure. Within outlier detection, Huang et al. (2013) leverage cumulative rank-based measures in a nearest neighbor algorithm, addressing limitations of density-based approaches, with Bhattacharya et al. (2015) extending this concept into a hybrid outlier detection method.

Rank-based anomaly detection methods utilize the ordering or ranking of data points rather than their absolute numerical values to identify anomalies. These methods are particularly useful in scenarios where traditional anomaly detection techniques based solely on numerical differences may overlook subtle but significant deviations from expected patterns. By focusing on the relative positions of data points within a dataset, rank-based approaches can effectively detect anomalies in various domains such as finance, healthcare, and network security. Unlike numerical analysis, which may be sensitive to outliers or noise, rank-based methods offer robustness against such disturbances, making them valuable tools for uncovering abnormal behavior patterns in complex datasets. Whether applied to identifying unusual stock market trends, irregularities in medical records, or anomalies in network traffic, rank-based anomaly detection methods provide a versatile and powerful means of enhancing anomaly detection capabilities and facilitating proactive risk management strategies.

Our approach utilizes the sum of absolute rank differences, akin to Manhattan distance or L1 norm of ranks. In a relevant research stream, Héberger (2010) introduces the concept of sum of ranking differences for comparing multiple methods in the case of a benchmark method. Héberger and Kollár-Hunek (2011) discuss the relevant ranking and validation procedures, and present its validation with a permutation test. Kollár-Hunek and Héberger (2013) extend it to cases with ties by using partial rankings. Héberger and Kollár-Hunek (2019) combine this with variance analysis to analyze the significance or randomness of differences. Brownfield and Kalivas (2017) use the sum of the ranking differences for the comparison of outlier detection methods. Previous studies have explored the concept of the sum of ranking differences primarily for method comparison; our manuscript distinguishes itself by constructing a rank-based

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anomaly detection framework. We present both normalized and unnormalized versions of the sum of absolute rank differences (SARD) curves for the practical visualization of anomalous patterns. In addition, our model computes feature importance scores, enhancing explainability by highlighting the variables that contribute the most to anomaly detection. Unlike prior work that typically focuses on a single characteristic, our approach is versatile and accommodates multiple attributes simultaneously.

The primary contribution of this paper lies in presenting a simple, yet practical, explainable rank-based anomaly detection method. We introduce the novel sum of absolute rank differences curve for visualizing outputs and a feature importance metric for enhanced explainability. The paper proceeds as follows. Section 2 presents the methodology, while Section 3 provides a demonstration of a simple example in addition to two practical applications with real-world data. Section 4 covers methodological extensions to handle different scenarios and Section 5 provides final remarks.

2 Methodology

This manuscript builds upon the concept of sum of absolute rank differences (SARD) to propose an extended rank-based anomaly detection framework that simultaneously considers multiple features and characteristics. We introduce both the normalized and unnormalized SARD curves (SARDCs), which provide visual diagnostics to illustrate the impact of varying feature rank differences. Additionally, we propose a relative importance score to enhance interpretability and explainability of the detected anomalies.

In this section, we present the relevant concepts and formulas underlying SARD, SARDCs, and feature importance that form the basis of our framework. Suppose we are interested in the difference between two groups X, the group of interest, and Z, the reference group. When a natural reference group is unavailable, a common approach is to use the average of all observations as a benchmark. This average reflects an aggregate expectation and can help assess deviations in the absence of a known reference, though it may be affected by bias and random error (Héberger and Kollár-Hunek, 2011). This manuscript uses averages in some examples, while the proposed framework is flexible and accommodates other types of benchmarks.

Let us assume both groups are observed over a set of *n* features, where each feature has *m* characteristics. Respectively, X_m^n (Z_m^n) corresponds to the mth characteristic of feature *n* in group X (Z). The topic of interest could be the comparison between the two groups with respect to the differences in rankings for the *m* characteristics over *n* features. In order to do that, denote $r(Y)$ as the rank of the random variable *Y* within a set. The rank $r(Y)$ is the rank of value *Y* within the set of values corresponding to the same characteristic across all features, where ranks are assigned in an ascending or a descending order given the context—e.g., for health care billing, lower values may receive lower ranks (i.e., rank 1 for the smallest value). Ties are handled using average ranks.

For example, in a health care billing example, if the number of prescriptions for five tests (X-ray, HIV, Blood, Urine, Chest) are 100, 70, 90, 120, and 50 respectively, then their corresponding ranks in ascending order (i.e., rank 1 assigned to the smallest value) would be 4, 2, 3, 5, and 1.

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This example illustrates a single characteristic (number of prescriptions) evaluated across five features (lab tests), where ranks are computed within the set of feature values for that characteristic. The complete worked-out example including two characteristics and all calculations is provided in Section 3.1 (see Tables 1 and 2).

Building on this intuition, we now formalize the Sum of Absolute Rank Differences (*SARD*) [1] metric to quantify the divergence in ranked values between provider $\mathcal{X}$ and reference group $\mathcal{Z}$:

$$SARD = \sum_{k=1}^{n}\sum_{j=1}^{m}\left|r(X_j^k) - r(Z_j^k)\right|. \quad (2.1)$$

To illustrate, consider a single feature *k*, and suppose we observe for characteristic $j=1$ that the ranks are $r(X_j^k)=4$ and $r(Z_j^k)=2$, and for characteristic $j=2$, $r(X_j^k)=3$ and $r(Z_j^k)=4$. These ranks correspond to values observed for the selected provider $\mathcal{X}$ and the reference group $\mathcal{Z}$, respectively. Then the total contribution to the SARD metric from this feature (i.e., the sum of absolute rank differences between $\mathcal{X}$ and $\mathcal{Z}$) is:

$$|4-2|+|3-4|=2+1=3.$$

One of the advantages of the proposed *SARD* metric via Equation (2.1) is that it is bounded and its upper and lower bounds are analytically available. The lower bound $\underline{SARD}$ is 0 corresponding to the case in which the rankings of all pairs coincide. The upper bound $\overline{SARD}$ is obtained when the orders are completely reversed as in:

$$\overline{SARD} = \begin{cases} mn^2/2 & n \text{ even} \\ m(n^2-1)/2 & n \text{ odd.} \end{cases} \quad (2.2)$$

Using the concepts introduced in Diaconis and Graham (1977), the proof of upper bound for SARD, $\overline{SARD}$, could be shown similar to Héberger (2010).

For a threshold-guided approach to flagging anomalous behavior, we can construct an empirical null distribution for the SARD statistic. Under the null hypothesis that a rank profile arises from the same underlying distribution as the benchmark, synthetic profiles are generated through repeated sampling and compared with the benchmark using the SARD metric. The 95th percentile of this null distribution can be used as a reference threshold to identify anomalies.

Next, we present the construction of the sum of absolute rank differences curve (SARDC). First, Equation (2.1) is rewritten as $SARD = \sum_{k=1}^{n}\Delta_k$, with Δ_k corresponding to the sum of absolute rank differences of the kth feature as in

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$$\Delta_k = \sum_{j=1}^{m} \left| r(X_j^k) - r(Z_j^k) \right|; k = 1, \ldots, n.$$

We now order the Δ_k 's in ascending order, obtaining $\Delta_{(1)} \leq \ldots \leq \Delta_{(n)}$. We define $S_0 = 0$ and $S_k = \sum_{i=1}^{k} \Delta^{(i)}$. Therefore, S_n coincides with SARD and S_k / S_n is the fraction of SARD due to the *k* ordered feature contributing the least to SARD. The curve connecting the points $(k/n, S_k / S_n), k = 0, \ldots, n$ is a convex, increasing function between the points $(0,0)$ and $(1,1)$. This normalized SARDC can be useful to visually assess if a given group, say X, has characteristics with very different rankings with regard to a reference group, say Y. In this case, it is important to look at the left derivative of the curve at 1, i.e., the slope of the last segment, since the larger the derivative, the greater the difference in the ranks of the characteristics of a group of the population. A look at the slopes of the nearby segments is also useful to detect if there are other groups with very different ranks.

Groups with the same ranking for all features and characteristics imply a straight horizontal line connecting the points $(0,0)$ and $(1,0)$ since $S_k = 0$ for all *k* features. On the other hand, the possible largest difference would coincide with the bisector of the normalized curve. Ekin et al. (2017) considered the area and the maximum distance (namely Gini and Pietra indices, respectively) between the curve and the bisector as a measure of the differences between two records. Here, we can consider the same quantities, but their interpretation is the opposite. If the SARDC is closer to the bisector line, that would correspond to a relatively equal contribution among features.

An unnormalized version of SARDC can also be obtained by connecting the points (k, S_k), $k = 0, \ldots, n$. In this case S_k depends on the actual values, with a value ranging between the lower bound of $\underline{SARD}$ (0) and its upper bound $\overline{SARD}$, as defined in equation (2.2). Such unnormalized curves could be useful, especially to see absolute differences between the group of interest and the reference group (X and Z respectively), as well as the distance from the absolute worst scenario, i.e., $S_n = \overline{SARD}$. See Figure 1 for an example demonstrating normalized and unnormalized SARDC curves.

An important aspect of the proposed framework is explainability, which is achieved by quantifying the relative importance of each feature within the sum of absolute rank differences. In order to do so, the relative importance score for k^{th} feature, $FI_k, k = 1, ..., n$, is calculated as

$$FI_k = \frac{\Delta_k}{SARD} \quad (2.3)$$

where $\sum_{k=1}^{n} FI_k = 1$. The feature importance score helps identify features that contribute the most in increasing the value of SARD. For example, in the case of a medical doctor's billing, this can

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indicate services or drugs that have been billed very differently when compared to a reference group of medical doctors.

The ranking procedure is straightforward and computationally efficient, with complexity $O(m \times n \log n)$ for *m* characteristics and *n* features. This complexity profile ensures the method remains tractable for high-dimensional problems. Feature importance scores, when desired, can be computed efficiently using the same ranked values without additional computational overhead.

3 Illustration

This section first demonstrates the proposed method using a simple health care billing example, followed by applications with two relatively large public real-world data sets. This analysis demonstrates the applicability of the proposed method in different domains, in addition to comparisons with similar applications. Codes and links to data are available upon request.

3.1 Simple Demonstration: Health care billing

First, we use a simple example to demonstrate the concepts. We compare the claim information of a selected medical provider (group X) with a reference group averaging the claim information of similar medical providers (group Z). There are five medical tests of interest ($n = 5$ features: X-ray, HIV, Blood, Urine, and Chest) and two characteristics of those medical tests ($m = 2$, number of prescriptions denoted by #, and average charges in 1000s USD denoted by $). Table 1 lists the raw data, whereas ranks of the tests with regards to the characteristics are presented in Table 2.

We obtain the following Δ_k values for all test types; $k = 1, ..., 5$:

$$\Delta_1 = |4-2| + |4-3| = \Delta_{(3)} = 3$$

$$\Delta_2 = |2-5| + |3-4| = \Delta_{(4)} = 4$$

$$\Delta_3 = |3-4| + |2-1| = \Delta_{(1)} = 2$$

$$\Delta_4 = |5-1| + |1-2| = \Delta_{(5)} = 5$$

$$\Delta_5 = |1-3| + |5-5| = \Delta_{(2)} = 2$$

In this example, the total SARD is $\text{SARD} = \sum_{k=1}^{5} \Delta_k = 16$. In the case of complete mismatch, the maximum possible value (i.e., upper bound) would have been $\overline{\text{SARD}} = 24$. Based on the observed rank differences, the feature with the highest relative feature importance score is

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”Urine” test, with $FI_{Urine} = 5/16 = 0.31$ followed by HIV tests ($FI_{HIV} = 4/16 = 0.25$). In other words, the urine test is the feature that contributes the most to the difference between X and Z when considering the number of prescriptions and the average charged amount.

Figure 1 shows the normalized and unnormalized curves for this example. To construct the curves, Δ_k ’s are sorted in ascending order as $(2,2,3,4,5)$ to construct S_k values as $(0,2,4,7,11,16)$, where $S_0 = 0$ by definition. The normalized curve on the left is formed connecting the points $(k/n, S_k / SARD)$ where $k = 0,\ldots,n$. In this plot, each point represents the fraction of SARD due to the *k* ordered feature, sorted from the least to the largest contribution, to form the increasing function between the points $(0,0)$ and $(1,1)$. Groups with similar ranking for all features and characteristics will form a curve closer to the bisector straight line. The unnormalized SARDC curve on the right panel is formed by connecting the points (k, S_k), $k = 0,\ldots,n$. The upper bound for the *y* axis of this plot is $\overline{SARD}$, the maximum possible value for *SARD* is calculated as $m(n^2-1)/2 = 2(5^2-1)/2 = 24$, based on equation 2.2 for an odd number of groups (i.e., $n = 5$). The group with the largest SARD value will form a curve that stands on top of all the other groups’ curves. Next, we provide a more comprehensive example of using SARD for anomaly detection with regards to medical prescriptions over time.

3.2 Health care billing over time

The percentage of overpayments within health care expenditures has increased over years, resulting in major financial losses and adverse health effects (Ekin et al., 2018; Ekin, 2019). Outlier detection methods such as univariate Lorenz curves (Ekin et al., 2017) and multivariate Gini index (Zafari and Ekin, 2019) have been used to reveal suspicious activities and flag potentially fraudulent providers. Most of the available methods have focused on numerical characteristics while ranks also could play an important role in decision making of auditors. The main interest in this study is the detection of doctors with anomalous billing information (with respect to their 5-year billing behavior) compared to doctors of the same medical specialty. Zafari et al. (2022) have approached this problem using weighted mixture of *m* Lorenz curves and by forming decision frontiers based on percentiles. However, that method may fail to capture some of the large ranking differences where SARD can provide insights in detecting anomalous prescription patterns.

The utilized dataset contains yearly billing ratios of 1,617 medical doctors across 30 drug groups from 2013 to 2017 as obtained in Zafari et al. (2022). These doctors correspond to 20 different medical specialties. Each doctor is compared to a reference group consisting of doctors with the similar medical specialty. In this analysis, we have $n = 30$ features (i.e., drug groups) captured over $m = 5$ characteristics (i.e., years). Table 3 shows the raw data for a given doctor. For example, this table show that in 2013, 0.1% of submitted prescriptions of this particular doctor is related to drug group 1 (i.e., G1). Our reference group for this doctor is formed by averaging billing ratios of drug groups for each year by considering billing information of all doctors with the same medical specialty.

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The proposed framework is applied separately to each medical specialty. The doctors with the highest SARD values could be used to generate potential audit leads. For each of these doctors with anomalous observations, we can compute the relative importance of the drug groups. For instance, the outlier doctor among cardiologists is found to be a doctor with the highest SARD value of 898.

In addition to identifying the anomalous observations, the proposed framework also identifies categories that have contributed the most to this behavior. Figure 2 displays the relative importance or contribution of each drug group in this calculation. As shown, drug group 5 has the highest feature importance score of $FI_5 = 74/898 = 0.082$, followed by drug groups 23 ($FI_{23} = 0.063$) and 15 ($FI_{15} = 0.06$). This helps auditors by identifying prescriptions and drug groups that separate this doctor from all other cardiologists.

Figure 3 presents the normalized and unnormalized SARDC of all the studied doctors on the left and right panels respectively. The blue dashed curves represent the doctor with the highest SARD value of 898. It should be noted that the y-axis on the right panel is constrained to [0,1000] as the actual SARD upper bound is relatively large, $5(30^2)/2 = 2250$, and its inclusion decreased the interpretative power. By referring to the unnormalized curve we can clearly see that this population has the highest SARD value and we can observe how the absolute rank differences are summed up to get to this value. In addition, the normalized SARDC displays the relative contribution of rank differences of the drug group. The closer this curve is to the bi-sector line, the more similar are the relative contributions. So for this particular doctor, not only he has the highest SARD value, his billing difference is similarly high for most of the drug groups. It is worth observing the complementarity of normalized and unnormalized curves: although normalization can be sometimes useful to have all the measurements on the same scale, the case of the doctor with SARD of 898 shows how the normalization might hide the anomalous behavior, as shown in the left side of Figure 3, while it is evident in the right side.

Next, we construct an empirical approximation of the null distribution for the SARD statistic to flagging anomalous prescribing behavior with respect to a threshold. Under the null hypothesis that a doctor's rank profile arises from the same underlying distribution as the benchmark, synthetic profiles are generated by sampling, for each drug group and year, a rank from the empirical distribution of doctors within the same specialty (e.g., cardiology in this example). Each resulting synthetic profile is then compared with the benchmark (the 5-year average profile) using the SARD metric. Applying this to our data, we repeated the sampling process 10,000 times, which yielded an empirical null distribution for the SARD values under the null hypothesis. The 95th percentile of this distribution, approximately 590 in the cardiology example, is used as a reference threshold. The observed SARD value of 898 substantially exceeds this threshold, falling in the extreme tail (greater than 99% of the bootstrapped null distribution). Figure 4 displays the empirical null distribution of SARD and the 95 % threshold.

In addition, an auditor can also compare different medical specialties in terms of overall SARD values of their corresponding doctors. For example, the differences of SARD across three of the medical specialties are displayed in Figure 5. The same analysis could be extended to all medical

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specialties to identify specialties which are more prone to higher differences in medical billing patterns.

Next, we compare these results with the overall Gini measures obtained in Zafari et al. (2022). Figure 6 presents a comparison of these two measures for the 70 cardiologists in the dataset. The complementary value of SARD can be recognized from this plot. We can see multiple patterns here, such as doctors with high values in terms of both Gini and SARD measures, or high on one and low on the other. For example, the cardiologist with the highest SARD value has a moderate Gini value compared to all other cardiologists (i.e., high SARD, low Gini). The cardiologist with the highest Gini value (shown with a red cross) ranks high in SARD with a quantile estimate of 0.92 (i.e., high SARD, high Gini). On the other hand, the cardiologist with the second highest Gini value (shown with a green triangle) ranks very low in terms of SARD with a quantile estimate of only 0.08 (i.e., low SARD, high Gini).

These differences highlight how SARD captures anomalies based on rank order differences, revealing patterns that traditional metrics like Gini might miss. SARD is particularly effective at detecting changes in the relative order of features, which can uncover subtle yet significant deviations not apparent when considering only the distribution of values.

Further analysis of the flagged doctors highlights the difference between the methods. For example, the doctor shown with a blue plus marker (highest SARD value and moderate Gini) has a billing pattern similar to the reference group (i.e., average pattern) for the doctor's top drug groups. However, there are substantial rank differences in their less commonly billed drug groups, which SARD effectively captures. This indicates that while the overall distribution of their billing amounts is not extremely unequal, the order in which they bill for certain drugs differs significantly from their peers. This kind of insight is crucial for identifying potential anomalies that may indicate irregularities or unusual behavior. Conversely, the doctor shown with a green triangle (very low SARD value and high Gini value) has significant differences in billing ratios for their top drug groups compared to the reference group. For instance, while the top prescribed drug group for cardiologists is group 24 with an average billing ratio of 68% over the five years, this doctor's top prescribed drug group is drug group 17 with an average billing ratio of 84% for the first four years and 1% in the last year, while the doctor's second prescribed group is group 24. This results in a high Gini value but a small rank difference, showcasing how SARD focuses on rank deviations rather than value distributions. These observations (high values in both measures, or high in one and low in the other) could often be important for investigations and audits. While the Gini index captures overall inequality in billing amounts, it does not reveal shifts in the relative ranking of procedures across providers. SARD identifies these rank-based deviations, flagging subtle anomalies that may indicate unusual or potentially fraudulent billing patterns. For example, a provider may bill common procedures at typical frequencies (yielding a moderate Gini score) while disproportionately favoring certain higher-cost services—patterns that SARD can uncover. By highlighting where and how behavior diverges from expected norms, SARD provides actionable, robust and interpretable insights that complement traditional metrics and support more targeted auditing decisions.

3.3 Hotel review ranking

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The impact of online word of mouth in hospitality management has long been demonstrated (Litvin et al., 2008; Pourfakhimi et al., 2020). In particular, understanding the impact of online user reviews on hotel room sales and evaluating the customer satisfaction is important (Ye et al., 2009). Since there are many reviews, hotel managers may want to focus on the most identifying ones, depending on their area of interest. This subsection presents an application of the proposed method to help the hotel managers in that pursuit.

The utilized data set is scraped from the Trip Advisor web site (Wang et al., 2010, 2011). It contains 675,069 reviews (written feedback) and stars/ratings submitted for seven criteria for 1,851 hotels across the world. In our analysis, we focus on numerical ratings for the seven criteria ($n = 7$ features: value, room, location, cleanliness, front desk, service, and business service), as shown in Table 4. Numerical ratings range from 1 to 5 stars. Given that we are only considering the ratings over these categories, the number of characteristics is considered to be 1 (i.e., $m = 1$). The raw dataset was pre-processed to remove reviews with missing (and negative) ratings. The final dataset has 81,085 reviews of 1,850 hotels. The number of unique reviews for each hotel ranges from 2 to 1,125 with median of 27 and average of 43.83.

The proposed method is flexible enough to offer various insights. For instance, hotel managers may want to identify reviewers submitting ratings that differ the most from other reviewers. While this could be straightforward for ratings submitted for a single criterion, it cannot be easily answered when considering ratings for multiple criteria. To do so, for a given hotel, we compute the average ratings over the seven criteria and use that as the reference group for ranking. The ranking of each reviewer is then compared with the ranking of the reference group. Reviewers with high SARD values are identified as those who submitted the most different ratings, considering all seven criteria. Table 5 displays the calculation for a selected reviewer. Row 1 of Table 5 lists the submitted ratings for a given user, while row 2 lists the average ratings submitted by all users for this hotel. Rows 3 and 4 show the ranks of the ratings for rows 1 and 2, respectively, and row 5 shows the absolute rank differences for each group. The summation of these rank differences gives us the SARD value of 15 for the given reviewer. This value is the highest estimated SARD among all the eighteen ratings submitted for this hotel. We can further identify the most influential criterion that makes this review different from others by computing relative feature importance scores. For this reviewer, service rating is found to be the most influential criterion $FI_{Service}$ score of 0.26 $(4/15)$. Analyzing the written feedback submitted along these ratings can help managers identify potential areas for further analysis and/or follow up with the reviewers.

Figure 7 displays the normalized (left panel) and unnormalized SARDC curves (right panel), where the blue curves represent the ratings of the outlier user. The upper bound for the y-axis of the right panel, $\overline{SARD}$, is calculated as $(7^2 - 1)/2 = 24$. Based on these curves, one can see how different this reviewer's ratings are compared to all 18 other reviews (right panel). The left panel shows that these differences are distributed consistently across the entire range of criteria, as indicated by the curve's steady deviation from the bisector.

Furthermore, there are other ways one can utilize SARD values in this context. For example, we can rank hotels based on the difference in their average ratings compared to all the hotels. In doing so, the reference group rankings can be computed based on the average ratings over all

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hotels or a group of similar hotels in terms of location, price range, etc. Each hotel's average rating is then ranked and compared with the reference rankings to calculate its SARD. The hotels with the highest SARD values are the ones that have received the most different ratings compared to all the hotels. This can help customers or hotel reservation websites identify the most different hotels.

In summary, traditional metrics like average ratings provide a broad view of customer sentiment but often miss subtle shifts in the relative ranking of features (e.g., cleanliness, service, amenities) across reviews. SARD detects these rank-based deviations, highlighting hotels or aspects of service that deviate from typical patterns. For instance, a hotel may have average ratings similar to peers but exhibit unusually high rankings for certain amenities while underperforming in others—signals that can indicate emerging issues or opportunities for targeted improvements. By revealing where and how customer feedback diverges from expected norms, our framework provides actionable insights for hotel management, enabling data-driven decisions to improve service quality, prioritize interventions, and enhance customer satisfaction. In short, anomaly detection in hotel reviews is crucial for trust, quality control, and business intelligence, ensuring the ratings are meaningful and actionable.

4 Methodological extensions

We have introduced the concepts of sum of absolute rank differences, SARDC and feature importance focusing on general cases. However, some of the practical needs may require extensions with respect to utility and consideration of ties. This section presents a brief overview of such cases with simple demonstrations.

So far, all the rank differences are assumed to have equal importance. For instance, the distance between ranks 1 and 2 is identical to the distance between ranks 4 and 5 in the example above. This assumption might not be acceptable in some situations; consider the 100 meters run in athletics at the Olympics where the difference between first and second runner (i.e. gold vs. silver medal) is much more important than the difference between sixth and seventh. At the same time, the utility of ranking first is much higher than ranking second or later and this difference should be taken into consideration. Therefore, for the latter aspect, it might be possible to consider a function U mapping from the set of ranks into an appropriate set. The previous demonstration assumed $U(x) = x$ for all ranks x. For illustrative purposes, we consider the previous example while applying the transformed function $U(x) = x^2$. The differences (now with superscript U) become:

$$\Delta_1^U = |U(4) - U(2)| + |U(4) - U(3)| = |16 - 4| + |16 - 9| = \Delta_{(3)}^U = 19$$

$$\Delta_2^U = |U(2) - U(5)| + |U(3) - U(4)| = |4 - 25| + |9 - 16| = \Delta_{(5)}^U = 28$$

$$\Delta_3^U = |U(3) - U(4)| + |U(2) - U(1)| = |9 - 16| + |4 - 1| = \Delta_{(2)}^U = 10$$

$$\Delta_4^U = |U(5) - U(1)| + |U(1) - U(2)| = |25 - 1| + |1 - 4| = \Delta_{(4)}^U = 27$$

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$$\Delta_5^U = |U(1) - U(3)| + |U(5) - U(5)| = |1-9| + |25-25| = \Delta_{(1)}^U = 8$$

Compared to the previous case, we can see that the first and the second ranked differences are now swapped, as well as the fourth and the fifth. In this case $SARD^U = 92$, whereas $\overline{SARD}^U = 144$.

Finally, one could consider ties in ranking. We use the widely utilized approach of keeping the total sum of absolute ranks constant, so that we assign, e.g., rank 1.5 for both tied at first, rank 2 if there are three ranked first, and so on. Let's demonstrate this using a classical example. The grass dataset (example 1 of section 5.7 from Conover (1998)) represents the rank preferences of twelve homeowners for four different types of grass planted in their gardens for a trial period. Each homeowner has ranked the four types as one of 1 (most preferred), 2, 3 and 4 (least preferred). Different studies have analyzed this dataset in terms of rank ordering of grass types (Taplin, 1997). In contrast, we focus on ranking homeowners based on their overall preferences and identifying the homeowner whose ranking is most different from the rest. This example has $n = 4$ features with $m = 1$ characteristic. The raw data (i.e., submitted ranks) with calculated SARDs are shown in Table 6.

Homeowners 7 and 8 with the highest SARD value are identified as the most different in ranking the grasses. This can be explained by the fact that the population prefers, on average, Grass 2 the most, followed by Grass 3, Grass 4 and Grass 1, whereas homeowner 7 prefers Grass 1, followed by Grass 3, Grass 2 and Grass 4.

5 Discussion

This article presents an explainable anomaly detection framework using the sum of absolute ranked differences. The proposed method can flag anomalous records, provide a graphical illustration, and quantify the contribution of each feature in detected anomalies in datasets characterized by multiple features. We have illustrated the method on real-world data sets from the hospitality and medical billing industries, and discussed its potential benefits.

The primary methodological contribution of this work is a rank-based anomaly detection framework that combines anomaly identification with explainable diagnostics across multiple features. Our framework is flexible for applied diagnostic settings, offering both visual and quantitative tools for understanding anomalies. Practitioners could use the 95th percentile of the empirical null distribution of SARD as a quantitative threshold for identifying potential anomalies. We show how SARD provides complementary insights to traditional metrics such as the Gini index, capturing rank-order anomalies that might be overlooked by distribution-based measures alone. Finally, our framework is modular in that extensions such as the use of utility-based weights allows practitioners to prioritize anomalies according to domain-specific factors such as severity, impact, or risk. This flexibility distinguishes our approach from existing rank-based methods, which typically focus on aggregate scoring or comparative performance. The proposed framework could also be extended to incorporate uncertainty making it adaptable to settings with noisy or incomplete measurements.

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In addition to applications in healthcare billing and hospitality reviews, the proposed rank-based anomaly framework is applicable in many other domains. For instance, in employee performance evaluation, if an employee consistently ranks much lower or higher than expected across multiple metrics, it could indicate an anomaly in performance that numerical analysis alone might miss. This could also apply to stock market analysis where a stock consistently ranks much lower in terms of price movement despite minor numerical differences. Rank-based anomaly detection could also be beneficial for website traffic monitoring, and credit risk assessment by capturing anomalies that may not be apparent through traditional numerical analysis alone.

While the proposed framework offers several advantages, certain limitations should be acknowledged. The additive nature of SARD does not account for potential dependencies among features, which may limit effectiveness in datasets with high multicollinearity. Additionally, the method assumes that rank-order differences are meaningful indicators of anomalies, which may not hold in all application domains. The use of large-scale rank data brings computational challenges due to the space of permutations (Diaconis, 1988). Therefore, most models in the literature have been relatively simple. While this helps with explainability, it may also limit the consideration of practical needs. The proposed ranking procedure itself remains computationally efficient and scales well with the number of features, since it relies primarily on sorting operations rather than combinatorial enumeration. For robustness analysis, while our applications use fixed observational data, future work could implement bootstrap resampling of rank assignments or introduce controlled perturbations to feature values to assess the stability of SARD scores and feature importance rankings across different scenarios. Future extensions could also incorporate correlation structures or domain-specific weighting schemes to address these considerations.

Several methodological extensions are worth considering. In addition to using different utility functions that incorporate the impact of ranks, other functions, such as squared rank differences, could be used to (de)-emphasize the distance between ranks. Recent explorations into transforming rankings into pairwise comparisons (Khetan and Oh, 2016a, b) have yielded insights into more tractable inferences for existing ranking models. These approaches could also be considered in relation to the proposed models in this manuscript.

Disclosure Statement

The authors report there are no competing interests to declare.

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Notes

[1] While the acronym SRD was used in the cited literature, e.g., Héberger (2010), we are referring to it as SARD to remark the presence of the absolute value of the rank differences.

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Figure 1: Left: normalized SARDC. Right: unnormalized SARDC

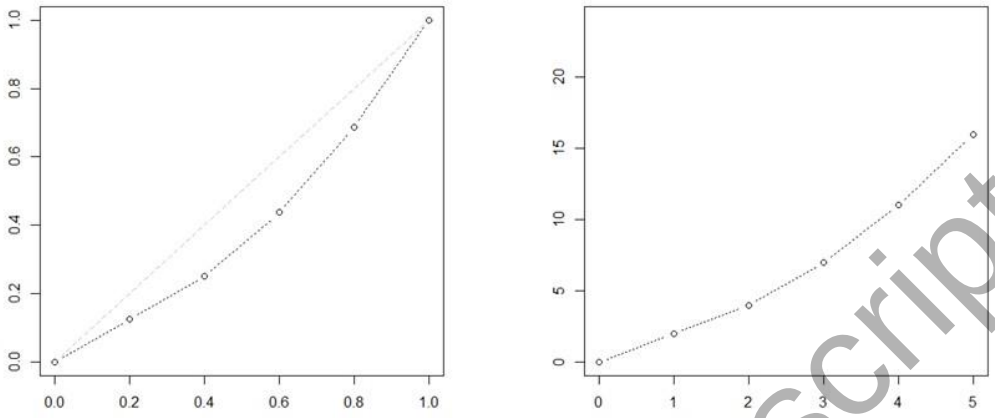

Figure 2: Relative importance of each drug group for the cardiologist of interest.

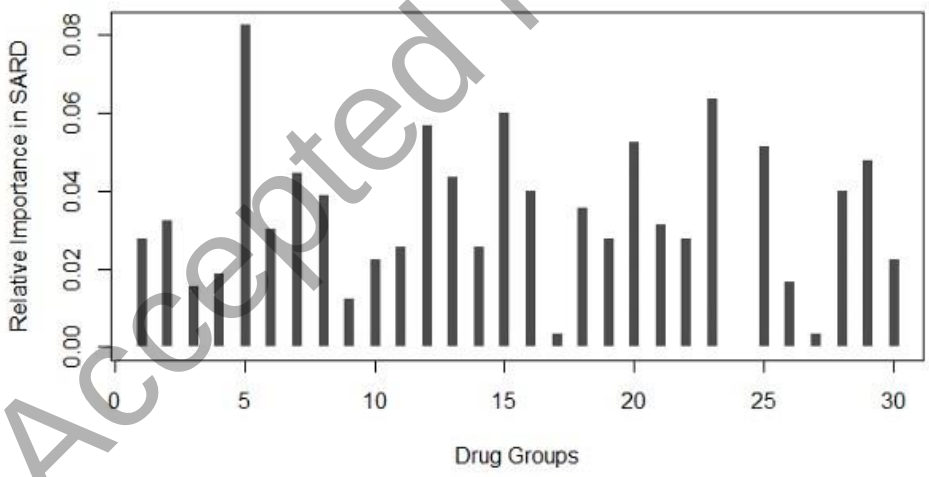

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Figure 3: Left: SARDC(normalized). Right: SARDC(unnormalized)

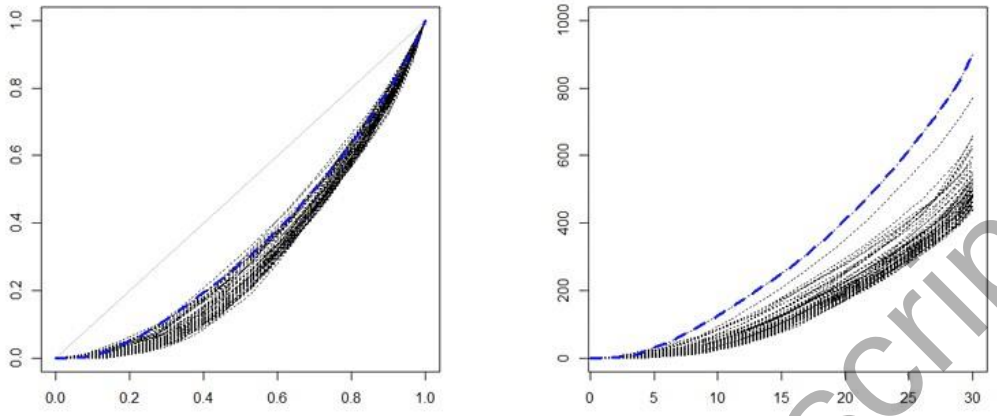

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Figure 4: Empirical distribution of SARD with the 95 % threshold.

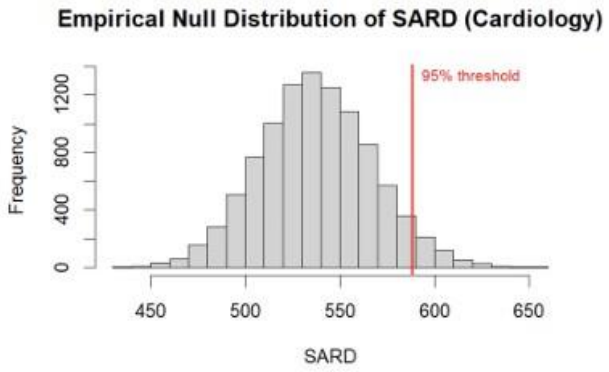

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Figure 5: Distributions of SARD across three different medical specialties.

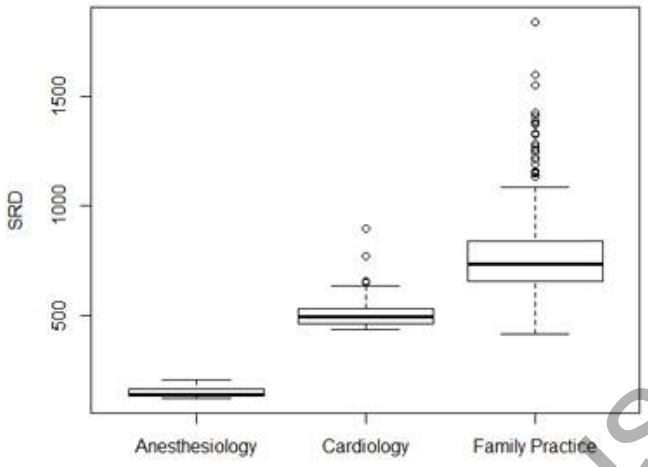

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Figure 6: Scatter plot of SARD versus Gini values for the cardiologists

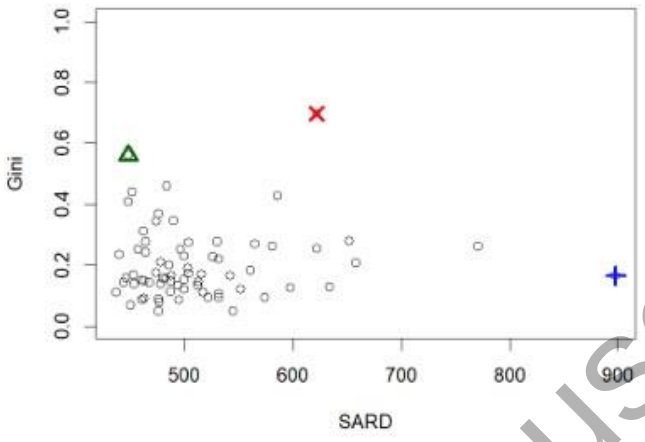

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Figure 7: Left: normalized SARDC. Right: unnormalized SARDC

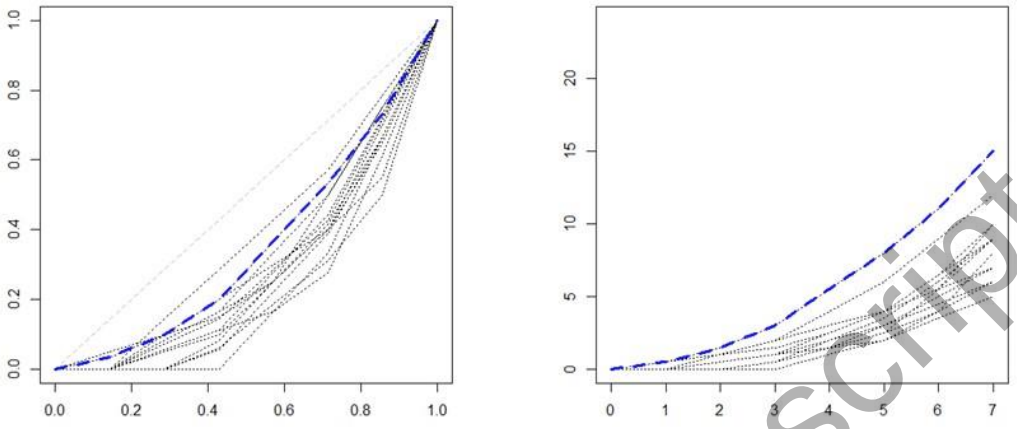

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Table 1: Number of prescriptions and average charges for five tests for groups X and Z

		X-ray	HIV	Blood	Urine	Chest
X	Number of prescriptions (#)	100	70	90	120	50
	Average charges ($)	20	15	10	5	70
Z	Number of prescriptions (#)	20	50	30	10	25
	Average charges ($)	70	100	20	50	150

Table 2: Ranks of number of prescriptions and average charges for five tests for groups X and Z

		X-ray	HIV	Blood	Urine	Chest
X	Number of prescriptions ranked (#)	4	2	3	5	1
	Average charges ranked ($)	4	3	2	1	5
Z	Number of prescriptions ranked (#)	2	5	4	1	3
	Average charges ranked ($)	3	4	1	2	5

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Table 3: Billing ratios of a given doctor for the 30 drug groups over the course of five years.

Doctor ID	Year	Specialty	G1	G2	G3	G4	G5	G6	G7	G8	G9	G10	...	G30
Doctor 1	2013	Family Practice	0.1%	0.0%	3.0%	23.8%	1.1%	0.0%	0.4%	0.3%	21.8%	5.1%	...	1.4%
Doctor 1	2014	Family Practice	1.6%	0.0%	0.5%	19.2%	0.2%	0.0%	0.3%	0.2%	16.2%	4.6%	...	6.2%
Doctor 1	2015	Family Practice	1.2%	4.0%	3.7%	19.4%	0.1%	0.0%	0.1%	0.2%	8.6%	6.1%	...	4.7%
Doctor 1	2016	Family Practice	2.0%	3.7%	1.8%	22.0%	0.1%	0.0%	0.2%	0.2%	15.7%	5.6%	...	2.6%
Doctor 1	2017	Family Practice	4.2%	4.9%	2.8%	25.6%	6.9%	0.0%	0.2%	0.2%	10.0%	4.2%	...	4.7%

Table 4: A random sample of the Trip Advisor review dataset for a particular hotel

Hotel ID	Hotel Location	User ID	Value	Rooms	Location	Cleanliness	Front Desk	Service	Business Service
100504	Seattle Washington	User 1	5	5	5	5	5	5	5
100504	Seattle Washington	User 2	2	3	3	5	2	2	3
100504	Seattle Washington	User 3	2	4	5	5	1	3	5
100504	Seattle Washington	User 4	5	5	5	5	5	4	3
100504	Seattle Washington	User 5	4	4	5	5	5	5	4

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Table 5: SARD calculations for a given review: Actual ratings (top), Ranking of the ratings (middle), and Absolute rank differences (bottom).

	Value	Rooms	Location	Cleanliness	Front Desk	Service	Business Service
Anomalous Reviewer Ratings	2	2	5	5	1	1	1
Hotel Average Ratings	3.05	2.38	4.83	3.16	3.27	3.44	2.33
Anomalous Reviewer Rating Ranks	3.5	3.5	1.5	1.5	6	6	6
Hotel Rating Ranks	5	6	1	4	3	2	7
Absolute Rank Differences	1.5	2.5	0.5	2.5	3	4	1

Table 6: Homeowner rankings of four grasses and their SARD values

Homeowner	Grass 1	Grass 2	Grass 3	Grass 4	SARD
1	4	3	2	1	4
2	4	2	3	1	4
3	3	1.5	1.5	4	3
4	3	1	2	4	2
5	4	2	1	3	2
6	2	2	2	4	4
7	1	3	2	4	6
8	2	4	1	3	6
9	3.5	1	2	3.5	1
10	4	1	3	2	2
11	4	2	3	1	4
12	3.5	1	2	3.5	1

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